

SHETH LUJ COLLEGE

PRACTICAL - 5

Aim:-Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

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CS_Practical5.py - D:/daniyal/CS_Practical5.py (3.14.6)
File Edit Format Run Options Window Help

# Diffie-Hellman Key Exchange Algorithm

p = 23
g = 5

print("Diffie-Hellman Key Exchange")
print("-----")

print("Public Prime (p):", p)
print("Public Base (g):", g)

# selects a private key
alice_private = int(input("\nEnter Alice's private key: "))

# selects a private key
bob_private = int(input("\nEnter Bob's private key: "))

# Calculate Alice's public key
alice_public = pow(g, alice_private, p)

# Calculate public key
bob_public = pow(g, bob_private, p)

print("\nAlice's Public Key:", alice_public)
print("\nBob's Public Key:", bob_public)

# Calculate shared secret key
alice_shared_secret = pow(bob_public, alice_private, p)
bob_shared_secret = pow(alice_public, bob_private, p)

print("\nAlice's Shared Secret Key:", alice_shared_secret)
print("\nBob's Shared Secret Key:", bob_shared_secret)

# Verify whether both keys are equal
if alice_shared_secret == bob_shared_secret:
    print("\nKey Exchange Successful!")
    print("\nBoth Alice and Bob have the same secret key.")
else:
    print("\nKey Exchange Failed!")

Ln: 38 Col: 57
```

OUTPUT

```
IDLE Shell 3.14.6
File Edit Shell Debug Options Window Help

Python 3.14.6 (tags/v3.14.6:c63aee6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>> ===== RESTART: D:/daniyal/CS_Practical5.py =====
Diffie-Hellman Key Exchange
-----
Public Prime (p): 23
Public Base (g): 5

Enter Alice's private key: 9
Enter Bob's private key: 5

Alice's Public Key: 11
Bob's Public Key: 20

Alice's Shared Secret Key: 5
Bob's Shared Secret Key: 5

Key Exchange Successful!
Both Alice and Bob have the same secret key.

>>>

Ln: 21 Col: 0
```

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